Birthday Paradox

- The birthday paradox (BP) states that you need far fewer people than expected to achieve a high probability of 2 sharing a birthday

- With only ~23 people there is already ~~a~~ > 50% of a match from 365 possible dates. This distinction is from pidgeon hole principle; paradox is about probability

- Birthday attacks apply this principle to hash functions and are an attack on collision resistance. Because hash functions produce a fixed ~~len~~ size digest regardless of input size, different messages can hash to the same value. An attacker exploits the birthday ~~bo~~ bound: finding a collision only requires approx. $2^{n/2}$ attempts for an n bit hash rather than 2^n.

Therefor the digest size must be large enough to make this statistically infeasible. SHA256 only offers 128-collision resistance despite its 256 output., so designers must account for this square root ~~relation~~ relationship